
PREDICTING TERM DEPOSIT SUBSCRIPTION USING DIRECT MARKETING CAMPAIGN DATA

December 7, 2024

Abstract

Banks have a wealth of customer data but often struggle to use it effectively for marketing. This study looks at a Portuguese bank's efforts to promote term deposits and aims to improve how it identifies potential subscribers. By analyzing a dataset with 45,211 customer records, including details about demographics, campaigns, and the economy, we develop a model to predict who is most likely to subscribe. The goal is to help the bank make better decisions, save resources, and connect with customers in a smarter and more personal way.

1. Introduction

Banks have access to extensive information about their customers, from basic details like age and occupation to more complex financial interactions. Despite this wealth of data, creating effective marketing strategies that resonate with customers remains a challenge. For products like term deposits, campaigns often yield low subscription rates, resulting in wasted effort and missed opportunities to connect with the right audience.

In this study, we examine the marketing challenges faced by a Portuguese bank, where only a small fraction of contacted clients opt to subscribe to term deposits. This challenge reflects a gap between the bank's marketing efforts and what customers truly seek, emphasizing the importance of adopting a data-driven strategy to better understand and reach potential subscribers.

By analyzing a dataset of over 45,000 customer records, we aim to uncover patterns in customer behavior and determine key factors influencing their decisions. The main objective of this project is to create a predictive model that enhances the effectiveness of marketing campaigns while ensuring they better address customer preferences. This method aims to make campaigns more targeted and meaningful, minimizing resource wastage and making stronger connections with customers.

2. Source

The data for this project is derived from a Portuguese bank's direct marketing campaigns. These campaigns were conducted through phone calls, with the goal of determining whether customers would subscribe to a term deposit. The dataset provides a comprehensive view of the interactions between the bank and its clients. This dataset, sourced from the UCI Machine Learning Repository¹ and is complete and does not have missing values. The demographic data includes attributes such as age, occupation, marital status, and education level, offering insights into the personal profiles of the customers. Campaign-specific information captures the frequency of contacts, the methods of communication, and the outcomes of previous interactions. Additionally, the dataset incorporates economic indicators like employment variation rates and interest rates during the campaign period, which are critical for understanding the external factors influencing customer decisions.

<https://archive.ics.uci.edu/dataset/222/bank+marketing> (Use 'bank-full.csv' from the zip file)

2.1 Description

The dataset provides the customers' information of the bank from May 2008 to November 2010. It contains 45,211 records with 17 attributes, described as follows in Table 1:

Table 1: Description of Input and Output Variables

Category	Variable Details
Client Demographics	
Age	Customer's age (numeric value)
Job	Customer's occupation type (categorical)
Marital Status	Marital condition of the customer (categorical)
Education Level	Highest education attained by the customer (categorical)
Credit Default	Indicates if the customer has any credit defaults (categorical)
Housing Loan	Indicates if the customer has a housing loan (categorical)
Personal Loan	Indicates if the customer has a personal loan (categorical)
Details of Last Campaign Contact	
Contact Method	Mode of communication used for the last contact (categorical)
Last Contact Month	Month when the last contact occurred (categorical)
Last Contact Day	Day of the week when the last contact occurred (categorical)
Contact Duration	Duration of the last contact in seconds (numeric)
Campaign Related Attributes	
Campaign Count	Total number of contacts made during this campaign for the customer (numeric)
Days Since Last Contact	Days elapsed since the customer was last contacted in a previous campaign (numeric)
Previous Contacts	Number of contacts made before the current campaign (numeric)
Previous Outcome	Result of the prior marketing campaign (categorical)
Target Variable	
Subscription	Indicates whether the customer has subscribed to a term deposit (binary: 'yes', 'no')

3. Exploratory Data Analysis

The dataset has no missing values across any column, ensuring data completeness for analysis.

3.1 Numerical Features

Key numerical features were analyzed to understand their distribution and variability:

- **Age:** Customers range from **18 to 95 years**, with an average age of **40.94 years**, reflecting a wide age group diversity.
- **Balance:** Account balances show significant variability, with values ranging from **-8,019 to 102,127** and an average of **1,362.27**. Negative balances suggest overdrafts, while outliers on the higher end could skew analysis.
- **Duration:** The mean call duration is **258.16 seconds**, with a maximum of **4,918 seconds**, indicating that call length may influence customer decisions.

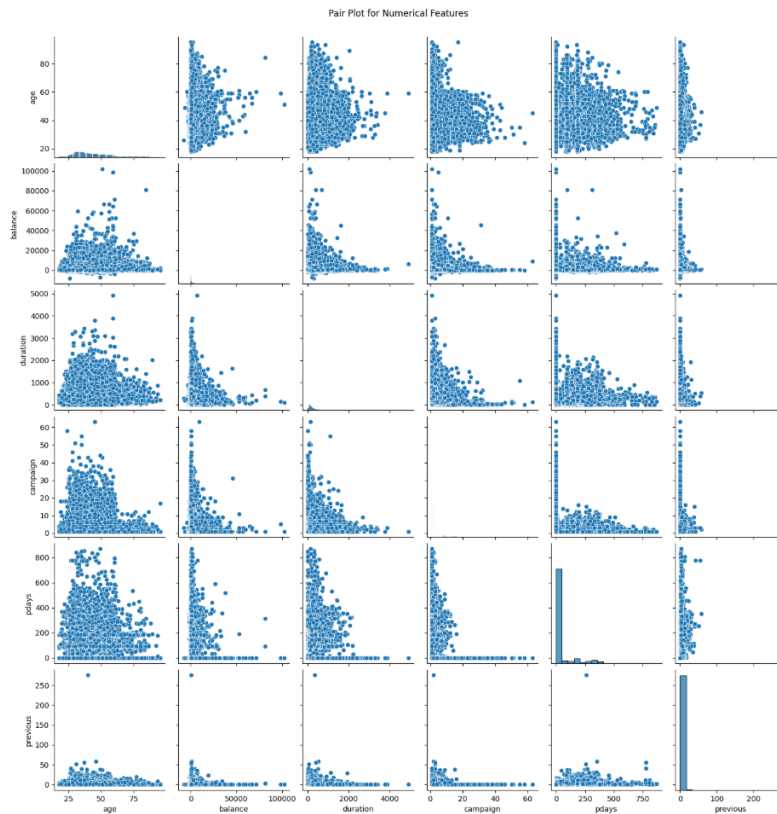
3.2 Numerical Features Analysis

To better understand the interrelationships and patterns among the numerical features in the dataset, a pair plot and a correlation heatmap were created. These visualizations provide insights into the structure of the data and help in identifying potential multicollinearity or non-linear relationships.

3.2.1 Pair Plot for Numerical Features

A pair plot of the numerical features (age, balance, duration, campaign, pdays, previous, and day) was generated to explore pairwise relationships among these variables. Observations include:

- Most features exhibit a right-skewed distribution, indicating potential outliers or concentration around lower values.
- Strong positive or negative linear relationships between variables are not visually evident.

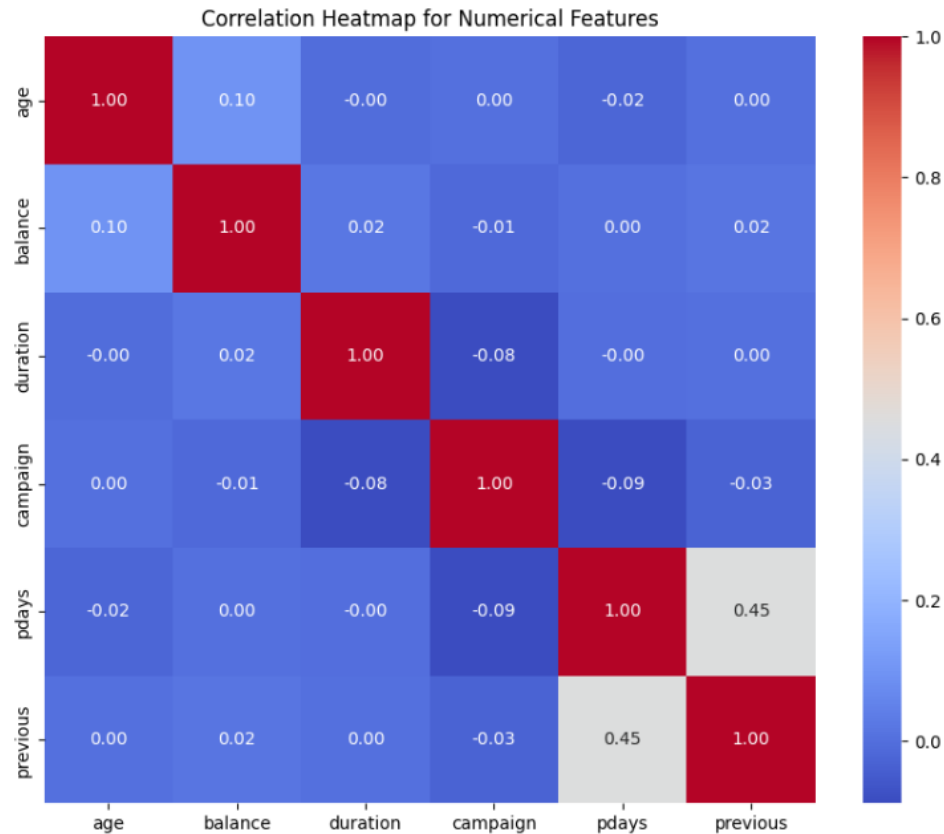


3.2.2 Correlation Heatmap for Numerical Features

A heatmap was generated to visualize the Pearson correlation coefficients between the numerical features. This highlights the strength and direction of linear relationships among these variables.

Key Insights:

- **Moderate correlation:** previous and pdays show a moderate positive correlation (0.45), suggesting they may convey overlapping information.
- **Low correlation:** Most features have weak or negligible correlations, indicating minimal multicollinearity among them.
- Features such as age, balance, and duration show minimal correlation with other variables, underscoring their potential as independent predictors.



3.2.3 Box Plot for Numerical Features

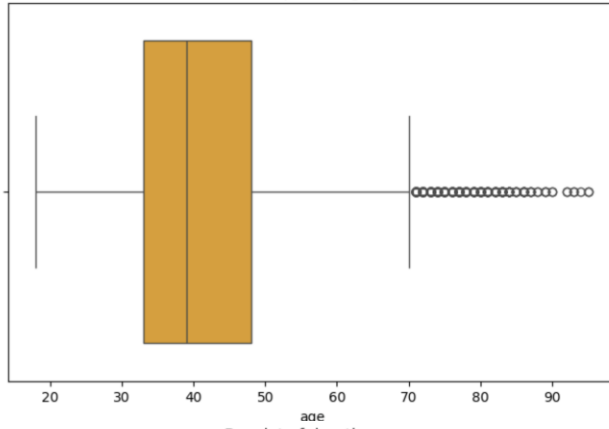
Box plots for each numerical feature were created to identify outliers and the range of data distribution.

1. **Age:** The distribution is concentrated between 30–50 years, with some outliers above 70.
2. **Balance:** Many outliers exist, with balances exceeding 60,000 euros.
3. **Duration:** The feature has a right-skewed distribution, with outliers above 3000 seconds.
4. **Campaign:** Most clients were contacted fewer than five times; outliers above 30 exist.
5. **Pdays:** Most clients have a value of -1 (not previously contacted), with some outliers reaching 800+ days.
6. **Previous:** While most values are 0, there are extreme outliers where clients were contacted more than 200 times.
7. **Day:** The features are uniformly distributed across all days of the month, with no evident outliers.

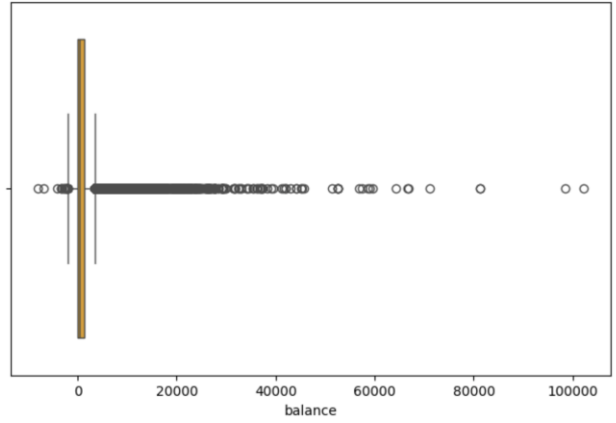
Insights:

- **Outliers:** Several features contain significant outliers, notably balance and duration. These outliers might influence model performance and may require scaling or removal.
- **Feature Relationships:** Weak correlations between most numerical features indicate minimal redundancy, suggesting that all features can be retained for initial modeling efforts.
- **Right-skewed Distributions:** Several numerical features, such as balance and duration, exhibit skewness, which might necessitate normalization for specific algorithms.

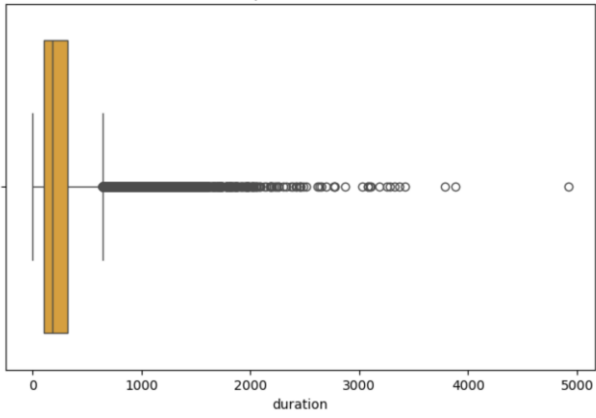
Boxplot of age



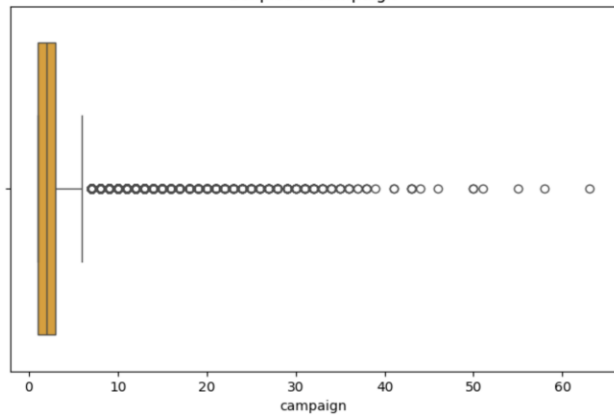
Boxplot of balance



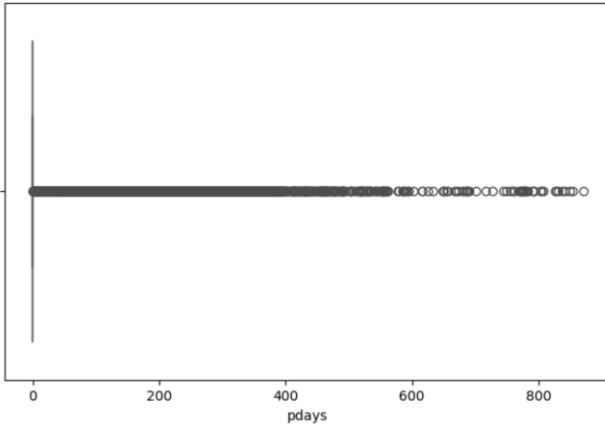
Boxplot of duration



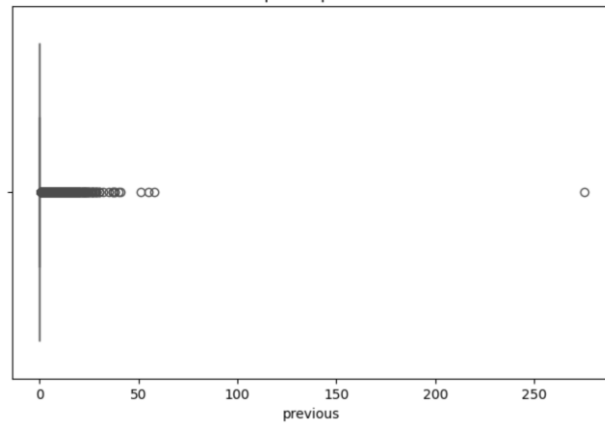
Boxplot of campaign

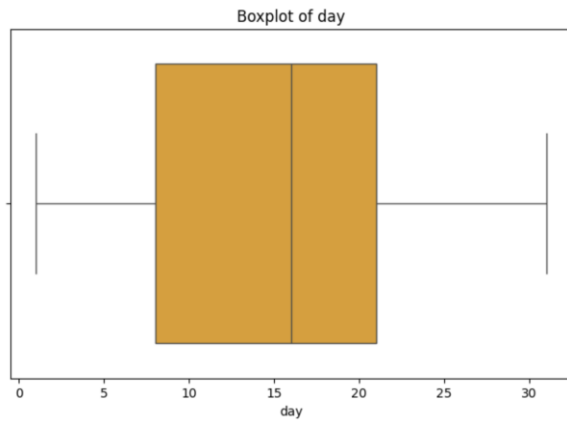


Boxplot of pdays



Boxplot of previous





3.3 Categorical Features

Categorical features reveal socio-economic patterns and campaign outreach details

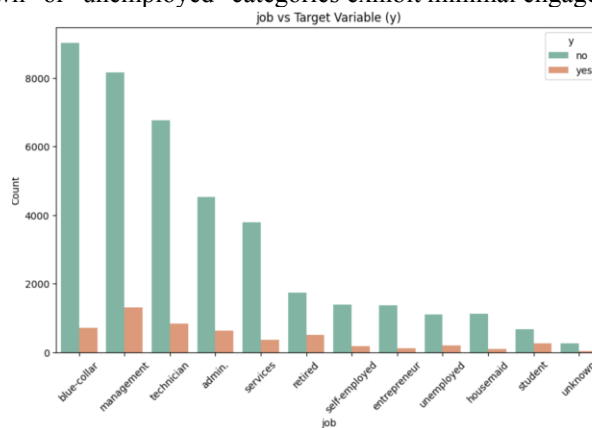
```
Summary statistics for categorical features:
      job marital education default housing loan contact \
count  45211  45211    45211   45211  45211  45211  45211
unique    12     3        4        2        2        2        3
top  blue-collar married  secondary    no    yes    no  cellular
freq    9732   27214    23202   44396   25130   37967   29285

      month poutcome      y
count  45211  45211  45211
unique    12     4        2
top     may  unknown    no
freq   13766   36959  39922
```

To understand the relationships between categorical features and the target variable (y), a series of bar plots were generated

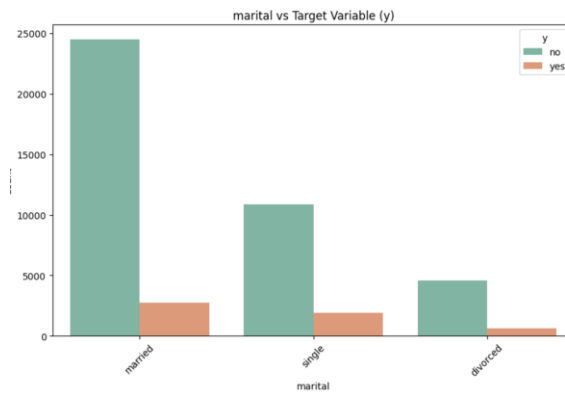
3.3.1 Job vs Target Variable

- The most frequent job category is "blue-collar," followed by "management" and "technician."
- "Management" and "technician" roles have a slightly higher proportion of yes responses compared to "blue-collar" roles.
- Customers in "unknown" or "unemployed" categories exhibit minimal engagement with the campaign.



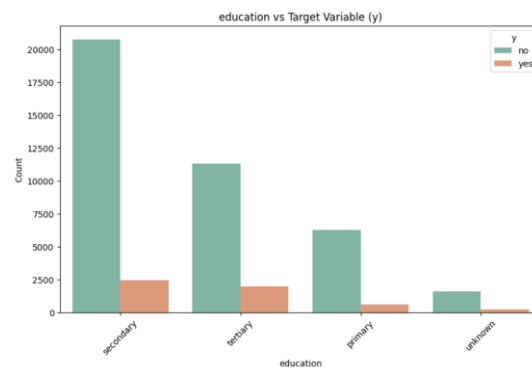
3.3.2 Marital Status vs Target Variable

- Most customers are married, followed by single and divorced.
- Single customers show a relatively higher proportion of yes responses compared to married or divorced individuals.
- This suggests that marital status could influence financial decision-making and receptiveness to marketing.



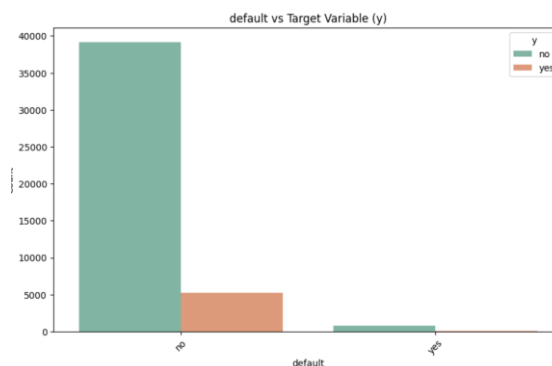
3.3.3 Education vs Target Variable

- Most customers have secondary education, followed by tertiary and primary education.
- Clients with tertiary education are more likely to subscribe (yes) compared to those with secondary or primary education.
- Education level may correlate with financial literacy and risk appetite.



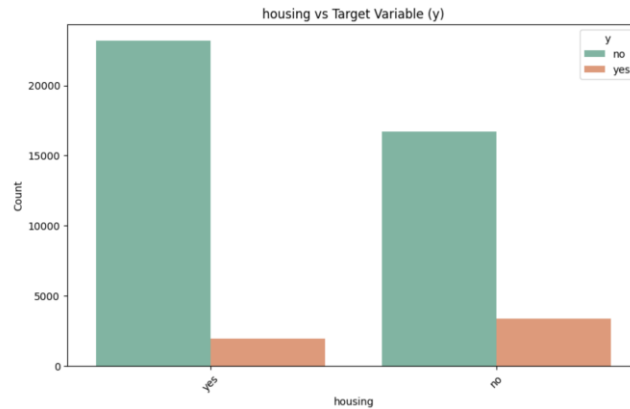
3.3.4 Default vs Target Variable

- Most customers do not have a default on credit (no).
- Customers with no history of defaults show a significantly higher likelihood of subscription.
- Default status is a strong indicator of financial stability and eligibility for new products.



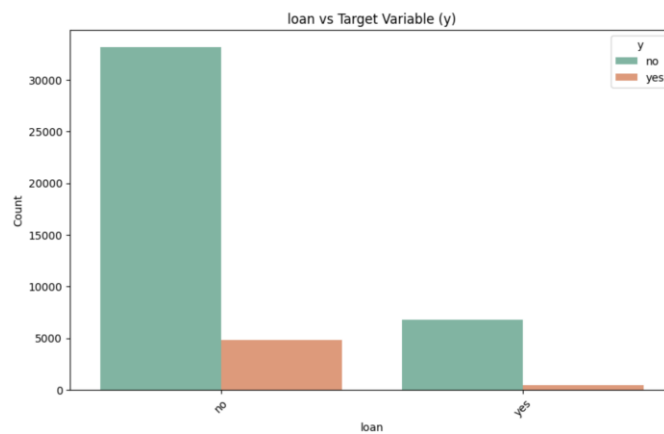
3.3.5 Housing Loan vs Target Variable

- Many customers have a housing loan (yes), while others do not.
- Customers without a housing loan are more likely to subscribe to a term deposit, indicating that financial obligations may reduce willingness to commit to additional investments.



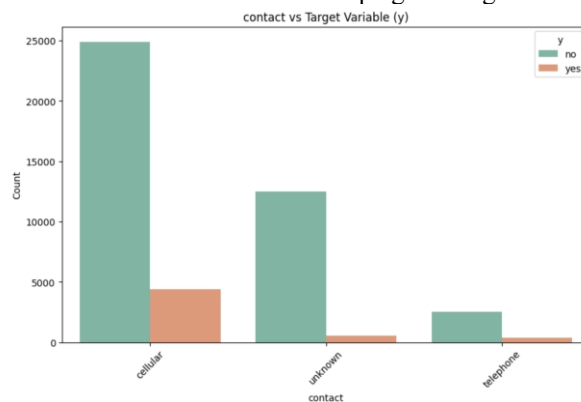
3.3.6 Personal Loan vs Target Variable

- Most customers do not have a personal loan (no).
- Like housing loans, customers without them are more inclined to subscribe, suggesting lower debt load may increase flexibility for savings products.



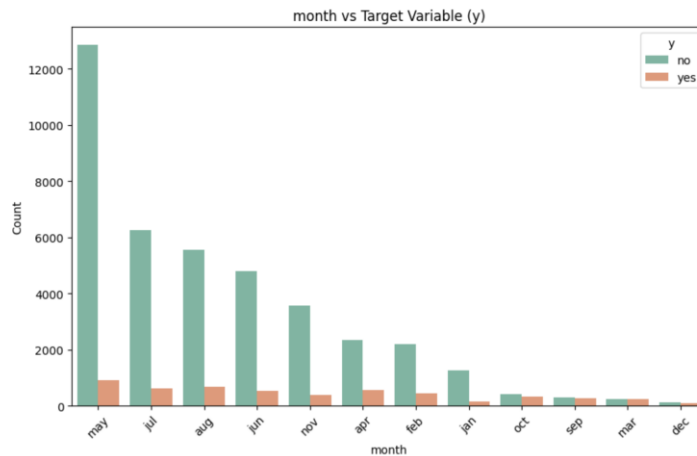
3.3.7 Contact Method vs Target Variable

- The most common method of contact is cellular, followed by telephone and unknown.
- Cellular contact is associated with a higher proportion of successful subscriptions (yes) compared to other methods.
- Contact method effectiveness could inform future campaign strategies.



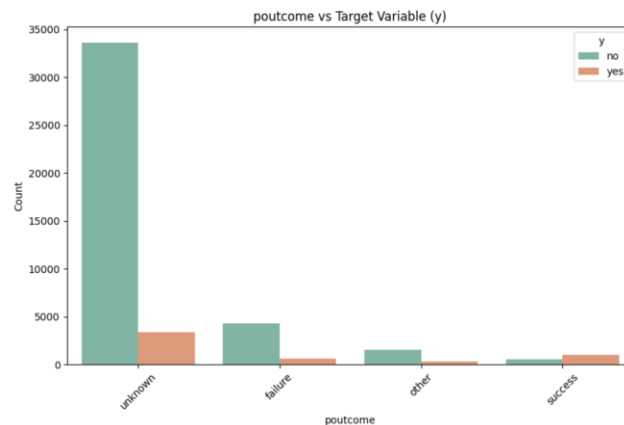
3.3.8 Month vs Target Variable

- Most contacts occurred in May, followed by July and August.
- Subscription rates (yes) tend to be higher during campaigns conducted in specific months, like March and October, indicating seasonal effects.



3.3.9 Previous Campaign Outcome (Poutcome) vs Target Variable

- Most customers fall under the "unknown" category for previous campaign outcomes.
- Customers with a "successful" outcome in previous campaigns are significantly more likely to subscribe again.
- Previous engagement history is a key predictor of future behavior.

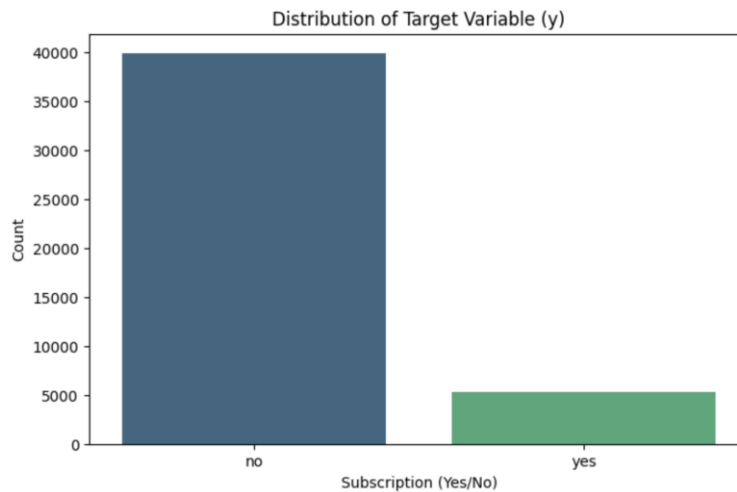


3.4 Target Variable Distribution

The target variable (y) is a binary indicator of whether a customer subscribed to a term deposit:

- **88.3%** of the customers did not subscribe ("no").
- Only **11.7%** subscribed ("yes").

This class imbalance suggests that the majority class ("no") dominates the dataset, which could bias predictions if not addressed.



4. Feature Engineering

Feature engineering is a critical step in any machine learning pipeline. It involves creating and transforming variables to improve model performance and ensure that meaningful patterns in the data are effectively captured. For this project, feature engineering was performed to address the specific challenges in the dataset, such as handling categorical variables, mitigating the impact of outliers, and capturing interactions between features.

4.1 Handling Categorical Variables

4.1.1 One-Hot Encoding:

Transformed categorical features such as job, marital, education, contact, month, and poutcome into one-hot encoded variables. For example, the job column was split into categories such as job_management, job_blue-collar, etc., to enable numerical representation for machine learning models. One-hot encoding is a method used to convert categorical variables into numerical form without incorrectly suggesting any inherent order or ranking among the categories.

4.1.2 Rare Category Grouping:

In the job column, rare categories with low frequencies (e.g., housemaid and unknown) were grouped under a new category called Other. This reduces noise and ensures that categories with very few samples do not disproportionately affect the model.

4.2. Interaction Features

4.2.1 Created Interaction Terms:

Interaction features were generated to capture relationships between multiple variables:

duration_campaign_interaction: Represents the product of duration (call duration) and campaign (number of contacts).

contact_poutcome_interaction: Combined the contact method and the outcome of the previous campaign (poutcome) into a single feature.

Interaction terms can reveal correlations or joint effects of multiple variables that may not be apparent when treated independently.

4.3. Binary Features

4.3.1 Derived Binary Flags:

long_campaign_contact: Indicates whether the client was contacted more than five times ($\text{campaign} > 5$).

recent_contact: Flags whether the client was contacted within the last 10 days ($\text{pdays} < 10$).

These binary flags simplify complex conditions into interpretable features, helping the model focus on specific thresholds.

	campaign	long_campaign_contact	pdays	recent_contact	\
0	1		0	-1	1
1	1		0	-1	1
2	1		0	-1	1
3	1		0	-1	1
4	1		0	-1	1
	duration_campaign_interaction		job_grouped		
0		261	management		
1		151	technician		
2		76	entrepreneur		
3		92	blue-collar		
4		198	Other		

The data indicates limited campaign exposure (single contact) and captures key interactions between call duration and contact frequency, while grouping rare job categories for better generalization in modeling.

5. Outlier Detection & Handling

Outlier handling was performed on the *balance* and *duration* features to mitigate the influence of extreme values and improve model robustness. *Balance* and *Duration* exhibited significant variability and outliers that could negatively impact model performance.

The Interquartile Range (IQR) method was used to detect outliers in the *balance* and *duration* features

Lower Bound= $Q1 - 1.5 \times \text{IQR}$

Upper Bound= $Q3 + 1.5 \times \text{IQR}$

Values below the lower bound or above the upper bound were flagged as outliers.

Applied Capping: Outliers were capped at the calculated lower and upper bounds:

balance: Accounts with unusually high or low yearly balances were clipped to fall within a reasonable range.

duration: Excessively long call durations were capped to remove extreme values while preserving meaningful variability.

Post-capping, the distributions of *balance* and *duration* became more uniform, improving both interpretability and model performance. The process ensured that extreme values did not bias predictions while retaining the meaningful variability in the data.

	balance	duration
count	45211.000000	45211.000000
mean	933.706974	234.939307
std	1176.765369	176.749932
min	-1962.000000	0.000000
25%	72.000000	103.000000
50%	448.000000	180.000000
75%	1428.000000	319.000000
max	3462.000000	643.000000

6. Feature Selection

Feature selection was conducted to identify the most relevant predictors for the target variable, optimizing model performance while reducing complexity and noise. The process incorporated statistical techniques, correlation analysis, recursive elimination, and model-based importance evaluation, ensuring that the retained features significantly contributed to the predictive task.

6.1 Correlation Analysis

Correlation analysis was performed on numerical features to evaluate their relationships with the target variable (y) and identify potential multicollinearity. Features such as duration and balance exhibited meaningful positive correlations with the target variable, indicating their predictive relevance. Conversely, features like campaign had weak negative correlations and were excluded due to their limited contribution. This analysis also highlighted that pdays and previous were moderately correlated, but only pdays was retained for its stronger association with the target.

6.2 Statistical Feature Scoring

Numerical features were assessed using SelectKBest with the ANOVA F-test to determine their statistical relevance to the target variable. Features such as duration, balance, and pdays emerged as significant predictors. For categorical features, the chi-square test identified strong dependencies between the target variable and features like job, education, and contact. These insights were important for the selection of features for further evaluation.

6.3 Recursive Feature Elimination

Recursive Feature Elimination (RFE) was applied using a Logistic Regression model to rank features based on their impact on classification performance. This iterative process identified duration, balance, recent_contact, and engineered interaction terms, such as duration_campaign_interaction, as crucial for accurate predictions. Features with minimal contributions were removed.

6.4 Random Forest Feature Importance

A Random Forest classifier was trained to evaluate feature importance, using its ability to capture nonlinear relationships and feature interactions. This analysis confirmed the significance of features such as duration and balance while also emphasizing variables like pdays and the binary flag recent_contact. Features like campaign and previous were ranked lower and later excluded from the final dataset.

6.5 Feature Engineering and Binning

The continuous variable age, which exhibited weak correlation with the target, was transformed into categorical bins (Youth, Adult, Middle-Aged, and Senior). This transformation revealed meaningful patterns in subscription behavior, with seniors demonstrating the highest subscription rates. Interaction terms and binary flags, such as long_campaign_contact and recent_contact, were also created to capture complex dependencies and improve interpretability.

6.6 Feature Elimination

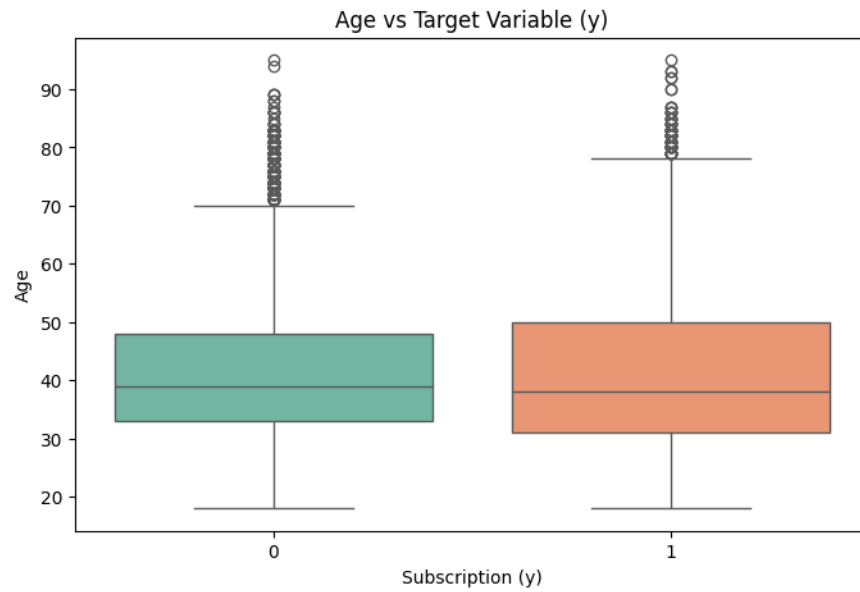
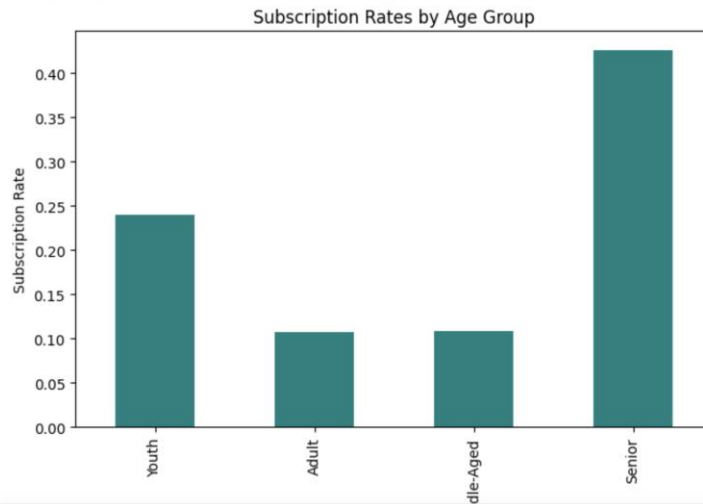
Features such as campaign and previous were excluded due to their weak statistical relationships and low importance in model-based evaluations. Additionally, the original age variable was replaced by its binned version to enhance interpretability.

```

age  age_group
0    58  Middle-Aged
1    44    Adult
2    33    Adult
3    47  Middle-Aged
4    33    Adult

Subscription rates by age group:
age_group
Youth      0.239521
Adult      0.107724
Middle-Aged 0.107980
Senior     0.426099
Name: y, dtype: float64
<ipython-input-51-30de19247c56>:6: FutureWarning: The default of observed=False is deprecated and will be
age_group_subscription = data.groupby('age_group')['y'].mean()

```



7. Data Splitting, Handling Imbalance, and Preprocessing

7.1 Final Dataset Composition:

The final dataset includes 44 features, combining selected numerical, one-hot encoded categorical, and engineered features. The inclusion of interaction terms (`duration_campaign_interaction`) and flags (`recent_contact`, `long_campaign_contact`) ensures that the dataset captures meaningful relationships and thresholds.

Updated feature names: ['job_blue-collar', 'job_entrepreneur', 'job_housemaid', 'job_management', 'job_retired', 'job_self-employed', 'job_services', 'job_student', 'job_technician', 'job_unemployed', 'job_unknown', 'marital_married', 'marital_single', 'education_secondary', 'education_tertiary', 'education_unknown', 'default_yes', 'housing_yes', 'loan_yes', 'contact_telephone', 'contact_unknown', 'month_aug', 'month_dec', 'month_feb', 'month_jan', 'month_jul', 'month_jun', 'month_mar', 'month_may', 'month_nov', 'month_oct', 'month_sep', 'poutcome_other', 'poutcome_success', 'poutcome_unknown', 'age_group_Middle-Aged', 'age_group_Senior', 'age_group_Youth', 'duration', 'balance', 'pdays', 'duration_campaign_interaction', 'long_campaign_contact', 'recent_contact']

7.2 Data Splitting

The dataset was divided into three subsets:

1. **Training and Validation Set:**
 - a. 85% of the data was used for training and validation.
 - b. Stratified splitting ensured the target variable (y) retained its original class distribution (88.3% for no, 11.7% for yes).
 - c. This split contained 38,429 rows.
2. **Hidden Test Set:**
 - a. 15% of the data was held out as a hidden test set to evaluate the model's final performance on unseen data.
 - b. This set contained 6,782 rows.
 - c. It was deliberately excluded from model training and intermediate evaluations to avoid data leakage.

7.3 Applying SMOTE for class imbalance:

In order to treat the imbalanced classes in the training set, Synthetic Minority Oversampling Technique, or SMOTE, was performed. This technique is used for creating synthetic examples for the minority class ($y=1$), so it balances both classes in the training set to 50%. This assures that the model will not be biased toward any class during training.

Class distribution in training set (after SMOTE):

```
y
0    0.5
1    0.5
Name: proportion, dtype: float64
```



7.4 Feature Normalization

Standardization was applied to numerical features to bring them onto a common scale:

StandardScaler: This scaler transformed the data by centering it around the mean (zero mean) and scaling it to unit variance.

Both the resampled training set and the validation set were scaled, ensuring consistency during model evaluation.

8. Machine learning Models

8.1 Logistic Regression

8.1.1 Model Implementation

A custom logistic regression model was developed using gradient descent optimization and L2 regularization. We performed hyperparameter tuning with various learning rates and lambda values to improve model performance. The best model was chosen based on its accuracy on the validation set. The training process was performed on the scaled and balanced dataset using the optimal hyperparameters identified during the tuning phase.

Learning Rate: 0.01
Regularization Parameter (lambda_): 0.01
Maximum Iterations: 1500

8.1.2 Evaluation Metrics

The model was evaluated on both the training and validation sets using the following metrics:

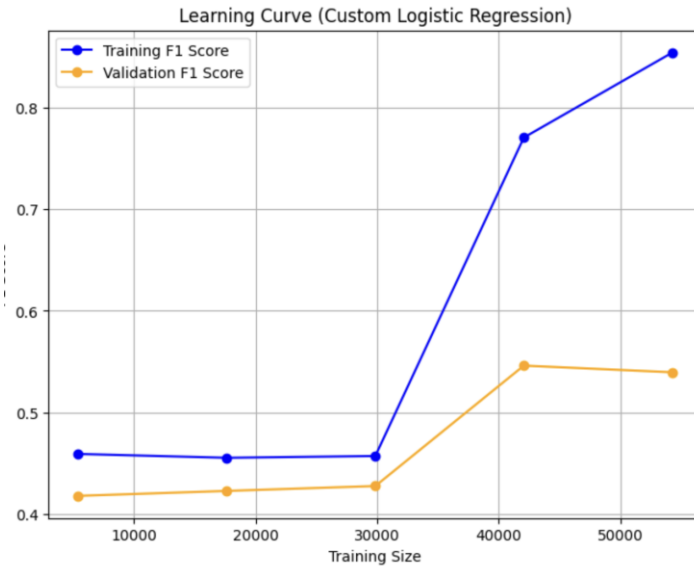
Set	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Training	85.31	84.68	86.23	85.45
Validation	83.58	40.01	80.87	82.23
Hidden Test	82.82	38.74	80.71	85.18

Generalization: The model generalizes well from training to validation and test sets, with only a slight decrease in accuracy and recall. This indicates minimal overfitting.
Class Imbalance : The lower precision across validation and test sets highlights challenges in managing the minority class. While SMOTE helped improve recall, it introduced some false positives, impacting precision.

8.1.3 Learning Curves

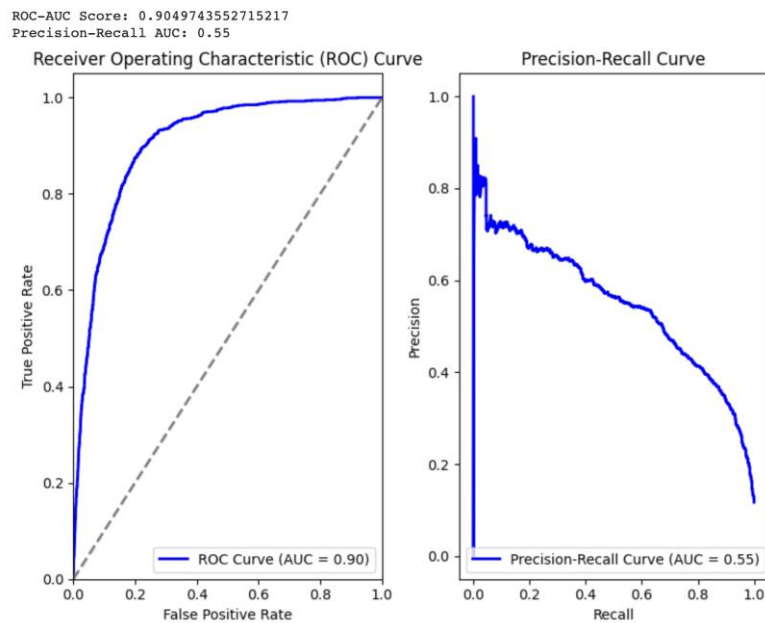
The learning curve provided insights into the model's bias-variance trade-off. Key observations include:

- **Training F1 Score:** The model achieved high F1 scores on the training set, steadily improving with larger subsets of training data. This indicates that the model learned effectively as more data was introduced.
- **Validation F1 Score:** The orange curve (validation F1 score) improves initially but plateaus as the training set size increases. This indicates that the model's generalization performance stabilizes with larger training data, without further gains in validation performance.
- The gap between the training and validation F1 scores indicates moderate variance, which can be addressed by additional regularization or hyperparameter tuning.



8.1.4 ROC-AUC and Precision-Recall Analysis

The ROC-AUC score for the validation set was 0.90, demonstrating strong discriminatory power between the positive and negative classes. The precision-recall curve showed a Precision-Recall AUC of 0.55, which highlights the challenge of balancing precision and recall in an imbalanced dataset.



8.1.5 Bias-Variance Trade-Off

Bias: The model exhibited low bias, as indicated by the high training accuracy and F1 score. The model effectively

captured the patterns within the training data.

Variance: Moderate variance was observed, as evidenced by the gap between training and validation F1 scores. While the validation performance stabilizes as more training data is added (indicating good generalization), there is still room to reduce variance through techniques like additional regularization, feature tuning, or hyperparameter optimization.

8.2 Naive Bayes

Naive Bayes is a probabilistic classification algorithm based on Bayes' theorem. It assumes that the features are conditionally independent given the class. The algorithm computes the posterior probability of a class C_k given a feature vector $x = x_1, x_2, \dots, x_n$ as:

$$P(C_k | x) = \frac{P(C_k) \prod_{i=1}^n P(x_i | C_k)}{P(x)}$$

Where:

- $P(C_k | x)$: Posterior probability
- $P(C_k)$: Prior probability
- $P(x_i | C_k)$: Likelihood of feature
- $P(x)$: evidence

The prediction is made by selecting the class that maximizes the posterior probability:

$$C^* = \arg \max_{C_k} P(C_k) \prod_{i=1}^n P(x_i | C_k)$$

8.2.1 Model Implementation

A custom Naive Bayes classifier was implemented using Laplace smoothing to handle cases with zero probabilities. The features were binarized to ensure compatibility with the Naive Bayes algorithm, which operates on categorical or binary inputs. The training process was conducted on the scaled, balanced dataset after applying SMOTE to address class imbalance. The implementation included:

- **Laplace Smoothing Parameter (α):** 1
- **Feature Binarization:** Applied to numerical features post-scaling to simplify the feature space for binary classification.

8.2.2 Evaluation Metrics

The Naive Bayes model's performance was evaluated on the training, validation, and hidden test datasets. The following table summarizes the results:

Set	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
-----	--------------	---------------	------------	--------------

Training	78.72	80.72	75.48	78.7
Validation	79.65	31.58	63.4	82.33
Hidden Test	79.33	31.22	63.81	82.1

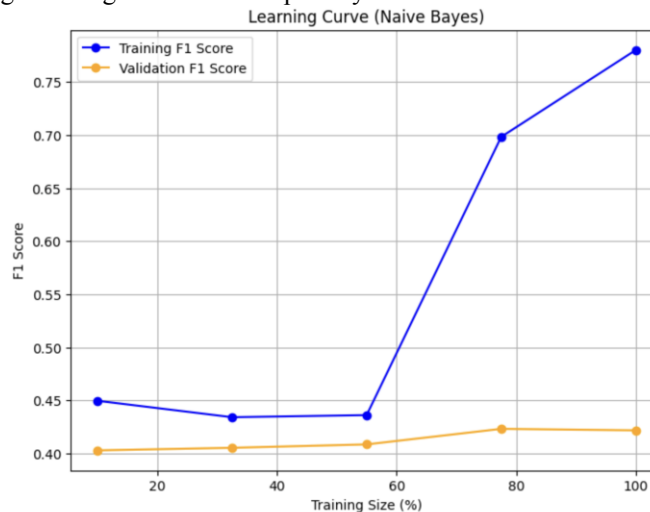
Training Performance: Naive Bayes achieved good precision and recall on the balanced training dataset, reflecting the model's ability to classify the resampled data effectively. The use of SMOTE ensured that both classes were represented equally during training, improving the model's recall for the minority class.

Validation and Hidden Test Performance: Despite SMOTE addressing the imbalance in the training set, the validation and test datasets retained their original class distributions, which impacted the model's precision for the minority class (positive class). This is a known limitation of Naive Bayes, as it assumes feature independence and is sensitive to imbalanced datasets in unseen data. While SMOTE improved recall for the minority class, precision remained low, reflecting the trade-off between these metrics.

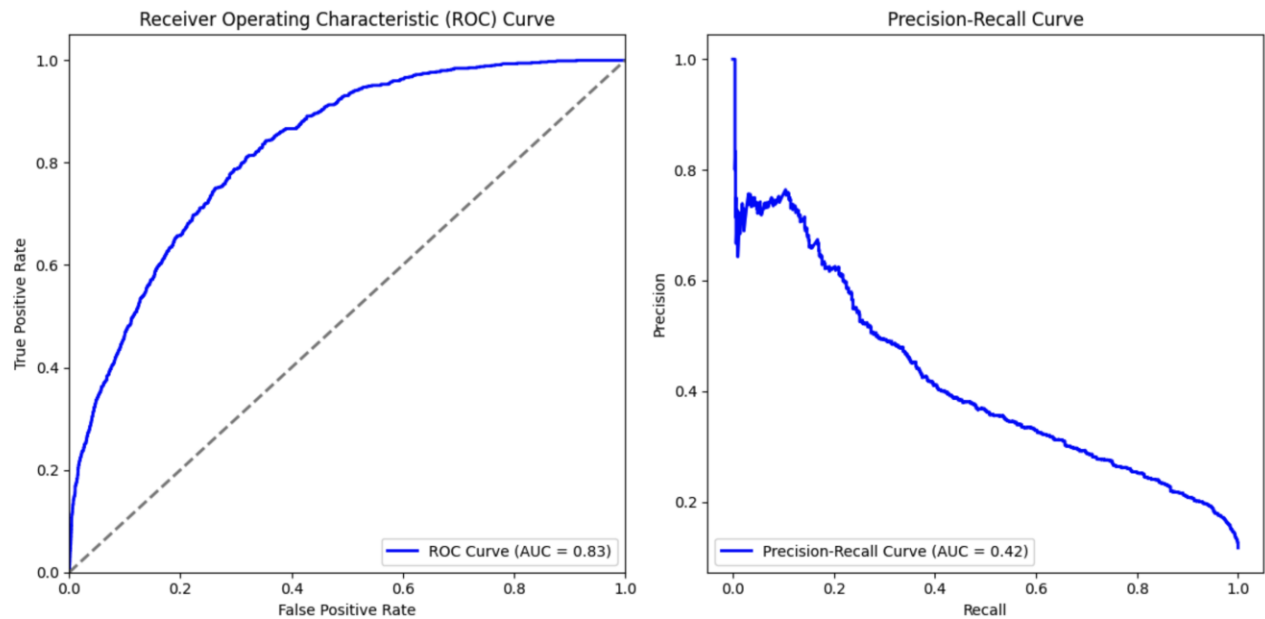
8.2.3 Learning Curves

The learning curves provided insights into the model's bias and variance:

- **Training F1 Score:** The model's training F1 score improved steadily as more data was added, demonstrating the model's ability to learn from the training data.
- **Validation F1 Score:** The validation F1 score increased initially but plateaued as the training set size increased, indicating limited generalization capability.



8.2.4 ROC-AUC and Precision-Recall Analysis



- **ROC Curve:** The model achieves good separation between classes. However, a higher false positive rate is observed, particularly for the minority class.
- **Precision-Recall Curve:** The steep decline in precision as recall increases shows the model's difficulty in maintaining precision while capturing more true positives for the minority class.

8.2.5 Bias-Variance Trade-Off

Naive Bayes is a simple model and trades off some accuracy due to its high bias and low variance, making it a good choice in many simple scenarios but less suitable for datasets with complex features. In this dataset which only contains 44 features, it is doable to use Naive Bayes. In our model:

- The model exhibited **moderate bias**, as reflected by its relatively lower performance on validation and hidden test sets.
- The gap between training and validation scores indicates **moderate variance**, due to Naive Bayes' assumptions of feature independence.

8.3 Neural Networks

A neural network consists of multiple layers of interconnected nodes, with each connection having a weight. The network is usually divided into three layers: an input layer, a hidden layer, and an output layer.

In a feedforward neural network, the output of each neuron in a layer is calculated as:

$$z_j = \sum_{i=1}^n w_{ij} x_i + b_j, a_j = \sigma(z_j)$$

Where:

- z_j : Weighted sum of inputs to neuron j.
- w_{ij} : Weight of the connection from input i to neuron j.

- x_i : Input i to the neuron.
- b_j : Bias term for neuron j .
- a_j : Activation output of neuron j .
- σ : Activation function.

The network learns by backpropagation by adjusting the weights and biases and uses the gradient of the loss function to minimize the prediction error.

8.3.1 Model Implementation

A feedforward neural network was implemented to classify the data. The model consisted of:

- **Input Layer:** Accepting the scaled features.
- **Hidden Layer:** One fully connected layer with 16 neurons and ReLU activation.
- **Output Layer:** A single neuron with a sigmoid activation function for binary classification.

The training process was performed on the scaled and balanced dataset using the following parameters:

- **Loss Function:** Binary Cross-Entropy Loss (BCEWithLogitsLoss).
- **Optimizer:** Adam Optimizer.
- **Learning Rate:** 0.001.
- **Batch Size:** 32.
- **Number of Epochs:** 10.

8.3.2 Evaluation Metrics

The model's performance was evaluated on the training, validation, and hidden test sets. Key metrics are summarized below:

Set	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Training	91.35	89.86	93.22	91.35
Validation	86.44	45.07	72.75	87.75
Hidden Test	86.08	44.35	74.78	87.53

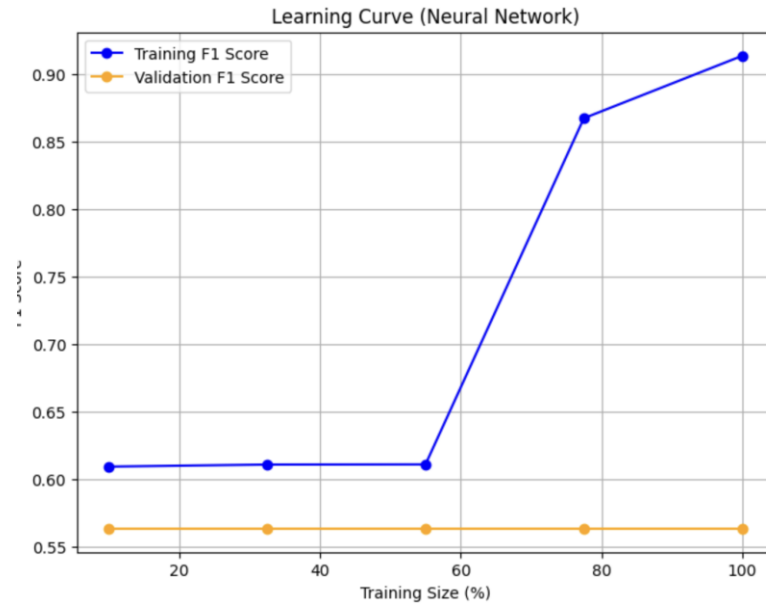
Training Performance: The model achieved high performance on the training set, indicating effective learning of the training data patterns.

Validation and Hidden Test Performance: Performance on the validation and hidden test sets showed slightly lower metrics, with a significant drop in precision for the minority class. This might be due to the imbalanced dataset.

8.3.3 Learning Curves

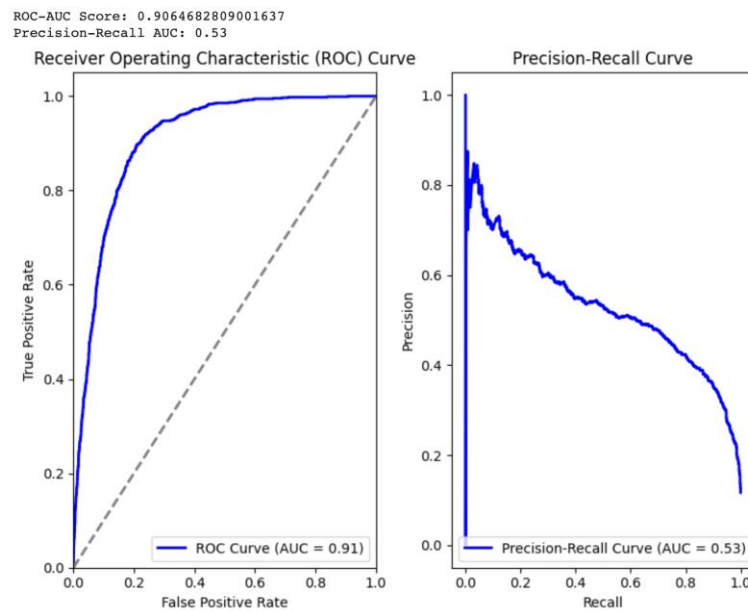
Training F1 Score: The training F1 score steadily improved as the model was exposed to more data, reaching a high value at the full dataset.

Validation F1 Score: The validation F1 score remained relatively flat, suggesting limited generalization improvement as the training set size increased. This may indicate overfitting or insufficient model complexity.



8.3.4 ROC-AUC and Precision-Recall Analysis

- The **ROC curve** shows a high true positive rate, though some false positives are introduced.
- The **Precision-Recall curve** highlights difficulties in maintaining high precision as recall increases, especially for the minority class.



8.3.5 Bias-Variance Trade-Off

Bias: The model exhibited low bias, as evidenced by the high training accuracy and F1 score.

Variance: Moderate variance was observed, with a noticeable gap between training and validation performance.

9. Results and Conclusion

The dataset originates from the banking and finance domain, specifically targeting clients during a direct marketing campaign. The goal is to optimize resource allocation by accurately predicting customer responses to marketing calls, thereby reducing costs and improving conversion rates.

Metrics to prioritize:

Recall for Class 1 (Minority Class):

- High recall is crucial because false negatives (missed potential customers) can lead to lost business opportunities.
- This is especially important in marketing campaigns where capturing interested customers is the primary objective.

Precision for Class 1 (Minority Class):

- Precision is important to minimize false positives, which can waste resources by contacting clients who are unlikely to subscribe.

F1-Score:

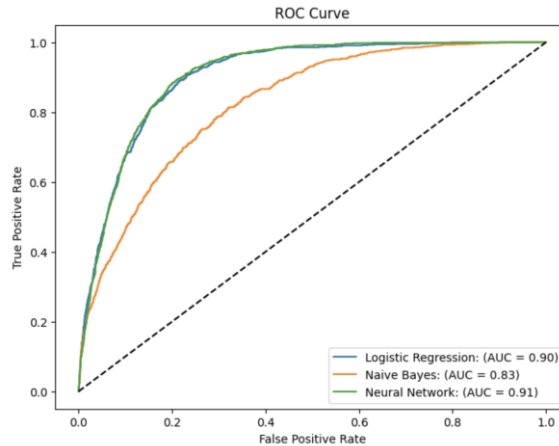
- As a balance between precision and recall, F1-score is a critical metric for overall model evaluation.

ROC-AUC:

- Measures the model's ability to distinguish between positive and negative classes across different thresholds, providing a holistic evaluation.

9.1 Key Metrics and Performance Comparison:

Metric	Logistic Regression	Naive Bayes	Neural Network
Training Accuracy (%)	85.31	78.72	91.34
Validation Accuracy (%)	83.58	79.65	86.44
Hidden Test Accuracy (%)	82.82	79.32	86.08
ROC-AUC (Validation)	0.90	0.83	0.91
Precision (Class 1)	40.01	31.58	45.07
Recall (Class 1)	80.87	63.40	72.75
F1-Score (Class 1)	82.23	82.33	87.75
Precision-Recall AUC	0.55	0.42	0.53



9.2 Model Insights

Neural Network:

The neural network performed the best in terms of accuracy, ROC-AUC, and F1-score, demonstrating its ability to capture complex patterns in the data.

However, the gap between training and validation scores suggests overfitting, indicating a need for stronger regularization or other improvements to enhance generalization.

Logistic Regression:

Logistic Regression showed good performance across all datasets, with high recall for the minority class and a competitive ROC-AUC of 0.90. Its simpler implementation and reduced risk of overfitting make it a practical choice for deployment in real-world scenarios.

Naive Bayes:

Naive Bayes achieved decent recall but struggled with precision and overall accuracy due to its independence assumption and sensitivity to imbalanced datasets.

9.3 Comparison with Baseline Model

Baseline (Logistic Regression):

Achieved strong overall performance with a validation ROC-AUC of 0.90, balancing recall (80.87%) and precision (40.01%).

Demonstrated generalization without significant overfitting.

Advanced Models:

Naive Bayes: While underperforming in precision and accuracy, it achieved competitive recall due to its reliance on conditional probabilities.

Neural Network: Excelled in F1-score and accuracy but exhibited overfitting due to a performance gap between training and validation datasets.

Conclusion:

The selection of models diverged from simpler, interpretable models to more complex architectures. Logistic Regression served as the baseline and demonstrated a competitive balance of interpretability and performance. Neural Networks, while the best-performing model in terms of metrics like F1-score and ROC-AUC, require further tuning to mitigate overfitting for practical deployment.

